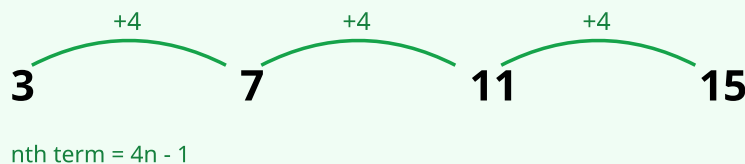


Geometric Sequences

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

- Adding instead of multiplying.
- Dividing the wrong way to find r .
- Decreasing sequences have a fractional ratio.
- Using r^n instead of $r^{(n-1)}$.

METHOD

- 1 Ratio $r = \text{term} \div \text{previous term}$
- 2 Check r is the same throughout
- 3 Next term = current $\times r$
- 4 nth term = first $\times r^{(n-1)}$
- 5 Check against the pattern

WORKED EXAMPLE

Continue 2, 6, 18, 54, ...
 $r = 6 \div 2 = 3$. Next = $54 \times 3 = 162$.
5th term = $2 \times 3^4 = 162$.

VISUAL AID

In a geometric sequence each term is the previous term multiplied by the same ratio r . With $r > 1$ the terms grow quickly; with $0 < r < 1$ they shrink towards zero.

1. (2 marks)

Write the next two terms: a) 3, 6, 12, 24, ... b) 1, 5, 25, 125, ...

2. (2 marks)

Find the common ratio: a) 4, 8, 16, ... b) 2, 6, 18, ...

3. (3 marks)

Write the next two terms: a) 5, 15, 45, ... b) 80, 40, 20, ... c) 1, -2, 4, -8, ...

4. (3 marks)

Find the term: a) first 3, $r = 2$, 4th term b) first 2, $r = 5$, 3rd term
c) first 100, $r = \frac{1}{2}$, 4th term

5. (3 marks)

Arithmetic or geometric (give d or r)? a) 2, 4, 6, 8 b) 2, 4, 8, 16 c) 5, 10, 20, 40

6. (3 marks)

Bacteria double each hour, start at 50. a) Number after 0, 1, 2, 3 hours. b) After 5 hours?

7. (4 marks)

Geometric, first term 4, ratio 3. a) First four terms. b) 6th term. c) Which term is 2916?

8. (5 marks)

Ball dropped from 20 m, bounces to 0.8 of previous height. a) Heights after 3 bounces. b) Arithmetic or geometric? c) Will it ever reach 0? Explain.